Real-Time MPC and Electricity-Price Forecasting Mathematical Formulations and Initial Experiments

PV–BESS energy-market research repository

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Abstract

This document describes the implemented real-time physical controllers for a PV plant with a shared battery and grid connection, and the design of the initial Italian electricity-price forecasting experiments. It distinguishes the shared open-loop ensemble plan, conditional adaptive recourse, the two-stage settlement price oracle, and ex-post realized accounting. It also states the feature and training constructions, empirical price scenarios, evaluation metrics, and native-resolution restrictions. Historical experiments are retrospective and are not presented as evidence that these policies or data were available at the time. Long-horizon controller results and final neural PV models remain pending.

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1 Scope, events and information

The system contains one PV plant and an AC-coupled battery behind a common point of interconnection (POI). There is no local load. The auction model participates in the day-ahead auction (DA) and up to three intraday auctions (IDA1–IDA3). Each signed order is assumed to execute fully. Upon submission its quantity is an immutable commitment; at publication its realized cashflow is booked once. An unpublished commitment is already fixed but not yet priced. The physical controller cannot change these submitted positions. Later auction decisions can change because the controller has changed the actual battery state.

A *continuous replay* carries the actual state, commitments and degradation memory chronologically across delivery days. It refers to repeated auction reoptimization and physical control, not a model of continuous order-book trading. Forecast objectives, value-to-go terms and risk penalties are planning quantities and are not booked as realized cash. The core auction formulation is in the companion `docs/mathematical_model.tex`; here we focus on the newer real-time models and the price inputs.

At each physical decision, the controller starts from measured or estimated battery energy and known commitments, obtains an eligible PV forecast and, when applicable, a settlement-price forecast, and solves over the remaining physical horizon. It implements only the first action and repeats at the next interval. Weather forecast cycles, lead times, their assumed release lag and source provenance are tracked; in the matched future comparison the newest eligible cycle covering the required interval is selected. Reviewed execution PV can contain synthetic replacements or configured night zeros, not solely independent meter observations. An oracle is explicitly noncausal when it uses realized future settlement or clearing prices.

2 Notation and shared physical constraints

Let $t = 0, \dots, T-1$ index the intervals remaining at one decision and Δ_t denote their duration in hours. A local day usually has 96 quarter-hour intervals, with 92 or 100 on daylight-saving transitions. Let $s \in \mathcal{S}$ index PV/price scenarios with $w_s \geq 0$ and $\sum_s w_s = 1$. All power is in MW, energy and commitments in MWh, and prices in EUR/MWh. A positive grid power denotes export.

Symbol	Meaning
$p_{s,t}$	Forecast available PV power in scenario s.
$c_{s,t}, d_{s,t}$	Battery charge and discharge power; their information ownership depends on the controller.
$u_{s,t} \in [0,1]$	Fraction of available PV used; the remainder is curtailed.
$e_{s,t}$	Battery energy at the interval boundary.
$g_{s,t}$	Net power exported through the POI.
q_t	Previously submitted net auction commitment for this interval.
$b^+_{s,t}, b^-_{s,t}$	Positive and negative deviations from the fixed commitment.
D_s, I_s, K_s, R_s	Pathwise wear, import cost, terminal inventory cost and settlement revenue.

For each path, the implementation enforces

$$g_{s,t} = p_{s,t}u_{s,t} + d_{s,t} - c_{s,t}, \qquad -P^{\text{imp}} \le g_{s,t} \le P^{\text{exp}}, \tag{1}$$

$$e_{s,t+1} = (1-\sigma)^{\Delta_t} e_{s,t} + \eta_c \Delta_t c_{s,t} - \frac{\Delta_t}{\eta_d} d_{s,t}, \qquad E^{\min} \le e_{s,t} \le E^{\max}, \tag{2}$$

$$0 \le c_{s,t} \le P^{\max} z_{s,t}, \qquad 0 \le d_{s,t} \le P^{\max}(1 - z_{s,t}), \quad z_{s,t} \in \{0,1\}, \tag{3}$$

$$b^+_{s,t} - b^-_{s,t} = \Delta_t g_{s,t} - q_t, \qquad b^+_{s,t}, b^-_{s,t} \ge 0. \tag{4}$$

Initial battery energy is fixed to the current state. POI bounds, energy bounds, optional terminal constraints and battery-mode exclusivity apply on every scenario. σ is hourly self-discharge and η_c, η_d are AC charge/discharge efficiencies. Any configurable import charge is an epigraph of $\max(-g_{s,t}, 0)$; the project does not impose an extra grid-usage fee merely because the battery imports.

With a throughput wear coefficient γ, one supported wear model is $D_s = \gamma \sum_t \Delta_t (c_{s,t} + d_{s,t})$. The alternative segmented Xu cycle-depth implementation includes its own segment-ordering and carried cycle memory; D_s below denotes whichever configured degradation model is used. See `src/optimization/degradation.py` and the companion mathematical model for its detailed constraints. A soft terminal target is, for example, $K_s = \rho |e_{s,T} - e^{\text{target}}|$; linear or configured piecewise-linear continuation values are alternative terminal treatments.

3 Common-control deterministic and ensemble MPC

3.1 Control ownership

The deterministic controller uses one forecast path. The ordinary ensemble MPC uses multiple PV paths but a *single open-loop physical schedule*:

$$(c_{s,t}, d_{s,t}, u_{s,t}, z_{s,t}, e_{s,t}) = (c_t, d_t, u_t, z_t, e_t) \quad \text{for all } s. \tag{5}$$

Grid injections and deviations still differ with $p_{s,t}$. All scenario POI constraints must hold. In a closed-loop replay the common schedule is not committed wholesale: only its first action is executed, and a new schedule is computed after the next observation.

For the price-blind penalty controller, define scenario-weighted absolute imbalance

$$A = \sum_s w_s \sum_t (b^+_{s,t} + b^-_{s,t}). \tag{6}$$

The solver minimizes

$$\kappa_{\text{blind}} A + D + I + K, \tag{7}$$

where D and K are shared trajectory costs and I is expected import cost. The penalty case in the research configuration uses $\kappa_{\text{blind}} = 500$ EUR/MWh. This is a trade-off between expected imbalance and other costs; it is not a lexicographic requirement that A attain its minimum. With a positive absolute-deviation penalty the split is tight at optimum, subject to solver numerical tolerances. Equal scenario/interval PV values may share identical interval-support variables; this exact deduplication does not delete members or relax the POI envelope.

3.2 Two-stage realized-settlement-price oracle

The optional oracle uses actual settlement curves, while its first stage remains price-blind. The stages are two optimizations within one MPC decision, not the DA and intraday auctions:

1. Minimize A subject to the shared physical constraints. Let U be the independently reconstructed feasible imbalance and L the solver lower bound.
2. If stage one meets the configured optimality criterion, or the explicitly configured time-limit fallback has a qualifying reconstructed relative gap, maximize settlement revenue less wear, import and terminal costs:

$$\max \quad R - D - I - K, \tag{8}$$
$$\text{subject to} \quad A \le U + \beta U + \delta, \tag{9}$$
$$R = \sum_s w_s \sum_t \left(\pi_t^+ b_{s,t}^+ - \pi_t^- b_{s,t}^-\right). \tag{10}$$

   Here β and δ are configured relative and absolute extra imbalance budgets. The second stage is restricted by the feasible stage-one incumbent, which need not be a proven minimum when the fallback is used.

For positive U, the reconstructed gap is $(U - L)/U$; the implementation also handles near-zero objectives explicitly. When enabled in selected studies, the fallback accepts a time-limited incumbent only if its gap is at most 1%. The standard relative solver MIP gap is 10^{-4} and no absolute MIP gap is configured. These studies allocate a separate 10-second optimization budget to each stage; building and validating models add wall time. A failed or invalid second-stage refinement retains the validated first-stage plan. A qualified incumbent is not described as a certified optimum. Direction-specific settlement prices require a binary sign restriction to prevent simultaneous artificial positive and negative deviations. The equal-price case uses signed settlement.

4 Forecast-economic and adaptive ensemble MPC

4.1 Scenario-based forecast economics

The research economic controllers use forecast, rather than realized, settlement prices $\hat{\pi}_{s,t}^+$ and $\hat{\pi}_{s,t}^-$. Let

$$R_s = \sum_t (\hat{\pi}_{s,t}^+ b_{s,t}^+ - \hat{\pi}_{s,t}^- b_{s,t}^-), \qquad L_s = D_s + I_s - R_s. \tag{11}$$

The implemented objective is

$$\min \sum_s w_s \left(D_s + I_s + K_s - R_s + \kappa_{\text{econ}} \sum_t (b_{s,t}^+ + b_{s,t}^-) \right) + \lambda \, \text{CVaR}_\alpha(L). \tag{12}$$

The economic imbalance coefficient is separate from the price-blind penalty and defaults to zero in these research cases. L_s excludes the artificial imbalance penalty and terminal salvage, and does not represent uncertain receipts on unpriced auction commitments. At the configured risk-neutral baseline $\lambda = 0$. When enabled, weighted monetary CVaR has the linear form

$$\text{CVaR}_\alpha(L) = \min_{\eta \in \mathbb{R},\, \xi_s \ge 0} \left\{ \eta + \frac{1}{1-\alpha} \sum_s w_s \xi_s : \xi_s \ge L_s - \eta \right\}. \tag{13}$$

The common-control economic ensemble controller uses the same physical ownership as the ordinary ensemble MPC. A deterministic economic controller has one PV path. Forecast settlement paths and PV members are combined under the declared scenario construction; the initial workflow does not learn a joint multi-auction/PV/settlement distribution.

4.2 Adaptive PV-history recourse

The adaptive controller optimizes objective (12) while allowing future controls to depend on disclosed PV histories. Let $n(s,t)$ be the node assigned to scenario s at time t. If two scenarios share a node then their charge, discharge, PV utilization and battery mode are identical:

$$n(s,t) = n(s',t) \Longrightarrow (c_{s,t}, d_{s,t}, u_{s,t}, z_{s,t}) = (c_{s',t}, d_{s',t}, u_{s',t}, z_{s',t}). \tag{14}$$

All scenarios share the first node and the current command. Nodes are formed from quantized observed PV prefixes up to a configured revelation lag; branch points occur only at configured future indices. They do not use future PV or a scenario label. Battery state follows each node's control history, and nodes do not recombine distinct prior battery histories. The current implementation does not reveal simulated settlement prices along a recourse path.

The prepared long study uses branching every four quarter-hour intervals, a one-interval observation lag and 0.1 MW PV-history bins; these are study settings, not structural constants. This is a restricted information policy, not perfect foresight. A retained representative path is an auction shadow projection and is not a prediction of which scenario will occur. Only the common first action is executed. Adaptive and ordinary ensemble models should be compared with matched objective, scenarios, information, state and physical settings before attributing a difference to recourse itself.

5 Execution, settlement and information comparisons

The grid-tracking baseline follows the committed grid position using its configured interval PV feedback and physical protection. MPC uses its own configured current-PV input, often the forecast. These information sets differ. Physical clipping or safety overrides and solver fallbacks are recorded.

For a realized interval, $\hat{g}_t$ is the executed grid power and $\hat{b}_t = \Delta_t \hat{g}_t - q_t$. Actual settlement is calculated from realized positive and negative deviations and realized settlement prices; actual auction cash is booked only on publication. Realized operating profit combines those cashflows and configured fees, imports and wear. It is not the optimizer's forecast objective. Since each policy carries its own state and can submit later auction orders, a profit comparison reflects the whole closed-loop policy. Ending SOC and terminal treatment must be reported alongside profit.

6 Initial price forecasting experiments

6.1 Data and feature construction

The initial retrospective experiments forecast the NORD DA, IDA1–IDA3 and imbalance-settlement prices. DA training starts in 2024 in the specified ridge study; its IDA and settlement training starts in 2025. A separate historical intraday experiment also used 2024 hourly exports as coarse labels expanded onto quarter-hours. Those repeated rows are not independent quarter-hour products. The newer search selects native hourly and native quarter-hourly products *before* any expansion and fits separate models for each market and resolution. Native price products are held constant within their duration, not linearly interpolated.

The initial price-only feature vector contains historical price lags and local calendar terms: sine/cosine representations of hour, weekday and annual phase, plus a weekend indicator. The European-context ridge study uses 24, 48, 168 and 336 hour lags, with configured same-clock fallback when a lag is unavailable. Its NORD fundamentals add zonal load, total-generation, solar and onshore-wind forecasts. Additional ablations add PSV gas, Milan/Turin temperature forecasts, French nuclear and hydro generation and France–Italy commercial exchanges, or Swiss

aggregate balance. A combined set includes these groups. An August-only set includes Swiss–Italy commercial schedules, directional *intraday* NTC and the Italian-NORD/Swiss spot spread. These are not physical border-flow measurements. No individual farm PV ensemble is a zonal-price feature.

The initial context run applies stated hypothetical publication policies: French/Swiss realized observations and PSV are shifted 48 hours; Swissgrid border observations use a nine-day shift with an assumed weekly release lag. A separate retrospective experiment uses previous-day realized information and a deliberately permissive source-day publication placeholder. Exact release clocks and revisions are not established by the download timestamp. Weather hindcasts are allowed in the stated retrospective temperature comparison. Missing source observations are retained as gaps, not converted to zero.

6.2 Ridge model, temporal split and residual paths

For training rows i, let x_i be the selected lag, calendar and optional fundamental features, y_i the observed price and $\tilde{x}_i$ the features standardized by training-set mean and scale. The fitted point model is

$$\hat{y}_i = \beta_0 + \tilde{x}_i^{\mathsf{T}}\beta, \qquad (\hat{\beta}_0, \hat{\beta}) = \arg\min_{\beta_0,\beta} \sum_{i \in \mathcal{T}_{\text{train}}} (y_i - \beta_0 - \tilde{x}_i^{\mathsf{T}}\beta)^2 + \alpha\|\beta\|_2^2. \tag{15}$$

The intercept is unpenalized. The initial European-context comparison sets $\alpha = 20$ and uses a 14-day chronological calibration tail. A predictor trained before the calibration period generates residuals on that period; the deployed point predictor is then refitted on all eligible labels through the fold cutoff. The calibration residuals do not come from the refitted model's in-sample predictions. Training labels and the feature-vintage eligibility rules are recorded separately from evaluation periods.

Whole complete local-day residual blocks $\varepsilon_{d,t}$ provide empirical price paths. For a selected calibration day d_j, a forecast sample is

$$\hat{\pi}_t^{(j)} = \hat{y}_t + \varepsilon_{d_j,t}. \tag{16}$$

Sampling a block preserves its within-day residual shape, rather than drawing independent noise at each interval. The configured DST policy controls which day lengths or local-clock slots can be used. Residual draws for separate days are independent; this does not model cross-day dependence or establish a joint PV–price distribution. The separate point forecast $\hat{y}_t$ need not equal the finite residual-scenario mean.

6.3 Scores and experimental design

For observed prices y_i and point forecasts $\hat{y}_i$, the reported absolute and relative errors include

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N} |y_i - \hat{y}_i|, \tag{17}$$

$$\text{WAPE}(\%) = 100\frac{\sum_i |y_i - \hat{y}_i|}{\sum_i |y_i|}, \tag{18}$$

$$\text{rMAE} = \frac{\sum_i |y_i - \hat{y}_i|}{\sum_i |y_i - \hat{y}_i^{\text{naive}}|}. \tag{19}$$

WAPE and rMAE are undefined when their respective denominators are zero. WAPE is a pooled normalization, not the mean of individual percentage errors; this avoids dividing separately by zero or near-zero prices. In the initial ridge report, MAE is averaged across complete daily scores,

while WAPE is pooled over the same common-day intervals; these need not weight 23/25-hour days identically. For forecast samples $x_1, \ldots, x_M$, the empirical CRPS at observation y is

$$\mathrm{CRPS} = \frac{1}{M}\sum_{j=1}^{M}|x_j - y| - \frac{1}{2M^2}\sum_{j,k=1}^{M}|x_j - x_k|. \tag{20}$$

CRPS evaluates the predictive distribution; it is not the ridge training loss or the eventual trading profit. The initial context ablation evaluates January and August 2026 and compares declared models on common complete delivery days. Feature groups can have different training coverage, particularly Swiss series and the 2026 border file; therefore shared test dates alone do not guarantee identical training samples. The results are development comparisons, not a full-year or universally causal test.

The expanded CPU search adds ridge, Lasso, elastic net, feed-forward neural networks, random forests, histogram gradient boosting, LightGBM and CatBoost candidates. These are point-model candidates on the library's interval features; the Lasso is not a reproduction of the full-curve LEAR benchmark. The first search validates separately by native resolution and reserves later dates for selection-fixed evaluation. Its quarter-hour later period includes August 2026, which was inspected in an earlier ridge study, so it is not globally untouched. Intraday and settlement searches are separate market tasks. The initial search is not an exhaustive hyperparameter search, nor are the selected candidates yet portable MPC forecast artifacts. The design draws on the benchmarking and metric discussion of Lago et al. [1]; model performance and forecast quality remain distinct from closed-loop trading value.

7 Prepared longer controller studies and limitations

The prepared annual study defines one continuous 2026 trajectory per farm and case; the ensemble study defines separate full January and August trajectories. The latter does not simulate the intervening months and therefore starts each month from its configured state. Both define matched pvlib and eventual neural forecast cases, with continuous commitments and latest-eligible forecast cycles. The present reviewed PV/measurement inputs end in August 2026, cached market series do not yet cover the whole year, and the available August ensemble lacks the preceding 31 July cycle needed by the first DA decision. The final neural models are being trained. Existing file presence is not proof of temporal coverage: complete replay preflight must check each required date before a run. No full-year or full-month controller result is claimed here.

8 Implementation references

- Common-control penalty and oracle: `src/optimization/mpc.py`; shared battery constraints: `src/optimization/physical.py`.
- Forecast-economic and adaptive model: `src/optimization/adaptive_mpc.py`; information nodes: `src/scenario_research.py`.
- Price features and fitting: `src/forecasting/energy_prices.py`, `src/forecasting/energy_price_features.py`, and `scripts/fit_energy_price_forecast.py`.
- Native-resolution model search: `scripts/search_energy_price_models.py`, `src/forecasting/price_search.py`, and `src/forecasting/price_search_models.py`.
- Experiment configurations and current input limitations: `configs/research/price_european_context.yaml` and `docs/reviews/long_horizon_2026_study.md`.

References

[1] J. Lago, G. Marcjasz, B. De Schutter and R. Weron, "Forecasting day-ahead electricity prices: A review of state-of-the-art algorithms, best practices and an open-access benchmark," *Applied Energy*, 293, 116983, 2021. https://doi.org/10.1016/j.apenergy.2021.116983.